
Shakespearean text generation

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Abstract

In this project, we make a Shakespearean text generator based on four recurrent neural network (RNN) architectures: vanilla RNN, 1-layer and 2-layer long short-term memory (LSTM) networks, and 2-layer gated recurrent unit (GRU). The models are trained on text from Shakespeare and *The Iliad* by Homer. In order to get the best model, a coarse-to-fine hyperparameter search and ablation are applied. To evaluate the synthesised text, we use both quantitative methods, like human evaluation and BERT score and qualitative methods, such as perplexity, spell percentage and Self-BLEU. Surprisingly, in our experiment, the more complex architectures LSTM and GRU did not dramatically outperform the simpler RNN. Increasing model depth did not lead to performance improvements. However, multi-domain learning proved effective. From a human evaluation perspective, it remained difficult to clearly distinguish which model performed best.

1 Introduction

This project studies how residual neural networks (RNNs) process sequential data for text synthesis and how the performance of RNNs differs between model architectures. Text synthesis is important for large language models (LLMs) in order to produce understandable text, both grammatically and factually. We have investigated training a 1-layer RNN, 1-layer LSTM, 2-layer LSTM, and 2-layer GRU on text by Shakespeare and *The Iliad* by Homer. We performed a coarse to fine search for hyperparameters, this included the number of hidden layers, neurons in the hidden layer, batch size, and learning rate. Further, we explored deeper LSTM architectures with and without dropout layers as regularization. Also, multi-domain learning ability was measured on a combined dataset of both Shakespeare and *The Iliad*. For evaluation metrics of our models we used a test dataset and measured its text synthesis ability based on perplexity, spelling, self-BLEU, and BERT score.

2 Related Work

Natural language processing (NLP) is an interdisciplinary subfield, which has components from major fields such as linguistics, computer science, and artificial intelligence. Text generation specifically has drawn tremendous attention after OpenAI published its chatbot ChatGPT. The development of neural network-based models has significantly advanced the capabilities of NLP systems. Among these models, Recurrent Neural Networks have played a fundamental role due to their ability to process sequential data and capture temporal dependencies within text.

Alex Sherstinsky's paper provides us a comprehensive tutorial on the fundamentals of Recurrent Neural Networks and Long Short-Term Memory (LSTM) networks. It rigorously derives the standard RNN architecture from a signal processing perspective and systematically discusses the unfolding of RNNs in time, highlighting key training challenges such as the vanishing and exploding gradient

problems. Building on these insights, the paper presents a detailed step-by-step development from the basic RNN to the vanilla LSTM architecture [1].

RNNs, by design, maintain a hidden state that evolves as new inputs are processed, which allows them to capture information about previous elements in a sequence. This sequential modeling capability makes them suitable for a range of natural language processing tasks, including language modeling, text generation, and sequence labeling. However, as pointed out in recent literature, standard RNNs are often limited by the vanishing and exploding gradient problem during backpropagation through time, which impairs their ability to learn long-range dependencies in language data [2].

To address these limitations, more advanced variants such as LSTM networks and Gated Recurrent Units (GRUs) have been proposed. LSTM networks incorporate memory cells and gating mechanisms which enable the network to selectively retain or discard information, effectively preserving relevant context over longer sequences. This property allows LSTMs to excel in tasks that require capturing long-term dependencies, such as machine translation, text summarization, and dialogue generation. GRUs, as a streamlined alternative, use update and reset gates to achieve similar objectives with fewer parameters, providing a computationally efficient approach while still addressing the challenges inherent in traditional RNNs [1], [3].

Both LSTM and GRU architectures have been widely adopted in modern NLP and have demonstrated strong empirical performance in a range of applications, including but not limited to language modeling, script generation, and question answering. Comparative studies, such as Mangal et al. (2019), have found that while both LSTM and GRU are effective for text generation, their relative performance may vary depending on the size of the dataset, the complexity of the task, and the required modeling depth [3].

3 Data

For our experiments, we use the Shakespeare dataset provided by TensorFlow, which contains the complete works of William Shakespeare in plain text format, [link here](#). Additionally, we experiment with a mixed dataset, for this we use the poem *The Iliad* by Homer, [link here](#). The Shakespeare text file (1.1MB) comprises approximately 1.1 million characters spanning plays, sonnets, and poems. *The Iliad* is in contrast a single poem with approximately 1 million characters. The data preprocessing we performed involved extracting all the unique characters which included letters, numbers, and special characters. The study used a character to number encoding and represented text as vectors of one-hot encoded numbers. All the text was processed together, first it was divided into sequences of 25 characters each, then two datasets were created, one for training input and one for ground truths targets. We partition the dataset into training (70%), validation (15%), and test (15%). During training we process data in batches so the last step of preprocessing is dividing the data into fixed size batches.

4 Methods

In order to evaluate the different RNNs we had to first create the data preprocessing and model architectures. Developing a strong baseline was needed to be able to compare the models against each other. We did a coarse to fine search for hyperparameters using the validation dataset then finally evaluated the trained models on the optimal hyperparameters on the test dataset and our other qualitative and quantitative evaluation metrics.

The preprocessing was done as in Section 3, the models were built using PyTorch due to its convenience as it has automatic differentiation, which means we can avoid computing the backward pass analytically. Additionally, it has the RNN, LSTM, and GRU layers builtin. All the models are character-level, processing one character at a time, with input and output size set to the number of unique characters in the data.

For the hyperparameter search we experimented with the nodes in the hidden states, batch size, learning rate, and number of hidden layers. Taking inspiration from Assignment 2 in the course we first performed a coarse search by applying random search over nearly one hundred parameter combinations, randomly sampling 25% of them to identify the best-performing combination.

Follow this, we conducted a fine search by making small adjustments to each parameter and again applied random search, sampling 25% of the new combinations to refine the results.

During the adjustment process, we manually set the value ranges for the number of nodes and batch size. For the learning rate, we first randomly selected six values within the range $[10^{-4}, 10^{-1}]$ in the coarse search, and then selected another six values from a narrower interval centered around the best result found during the coarse search.

4.1 Quantitative evaluation

Human Evaluation: The human evaluation on the model output text includes the following aspect:

- Semantic evaluation: Evaluate whether the text is readable and understandable.
- Grammar evaluation: Evaluate Whether the text uses correct grammar.
- Shakespeare similarity: Evaluate whether the text generates like Shakespeare.

BERT score: The BERT score mimics human evaluation on generated text. The idea behind is using BERT model to calculate semantic similarities between reference and generated texts. Due to the dataset limitation, the BERT score will not regularized based on modern english semantic. The score is calculated based on the BERT score package in python [4], [5].

4.2 Qualitative evaluation

Perplexity: Perplexity aims to measure the certainty when the model writes next word/character. The value is calculated based on the cross entropy loss from the model, which is the summation of all true log probabilities when the model predicts correct character. The log probabilities can be obtained through softmax function of the model output score. The lower the perplexity, the better the model generates the next character [6], [7].

Spell Percentage: This metrics measures the correctness of a model produce, it takes reference on pyspellchecker and common contractions. This percentage requires a completeness on a single word [8].

Self-BLEU: Self-BLEU score is to measure the diversity of wording in text by comparing every n contiguous sequence with reference text. Since BLEU is originally used to determine the similarity between two sentences, it can similarly evaluate how closely one generated sentence matches the others within a set. By treating one sentence as the hypothesis and the remaining sentences as references, we compute BLEU scores for each sentence individually, then average these scores to obtain the document’s Self-BLEU. For our model with output of 1000 characters, we choose to use 4-gram Self-BLEU score. A higher Self-BLEU indicates lower diversity among the generated sentences [6], [9].

4.3 Text synthesis

Text synthesis: This function generates a sequence of text by predicting one character at a time using a trained model. It starts with an input string, converts it into one-hot encoded vectors, and feeds them into the model along with the previous hidden states. At each step, it uses the model’s output to predict the next character by sampling from a probability distribution (adjusted by temperature) or using nucleus sampling. It updates the input with the new character and repeats the process until the text length is reached.

Nucleus sampling: This sampling method applied in the experiment was introduced by Ari Holtzman et al., ICLR 2020[6]. It selects the smallest set (‘nucleus’) of highest-probability tokens whose cumulative probability exceeds a manually set threshold p and then samples from that reduced subset. Ensuring sampling is from the most “core” of the distribution while retaining randomness.

Temperature scales the model’s output logits, so that it controls the "creativity" of our text generator. Text generated at low temperature(e.g 0.2) tends to be stable and coherent, even repetitive, while at high temperature(e.g 1) the result will be rare and creative but unexpected and incoherent.

4.4 Other

Mixed datasets: For training the models on a mixed dataset we compared the Shakespeare results to only training on *The Iliad*, and finally combining these text so that the training data was 75% Shakespeare and 25% *The Iliad* and then only evaluating it on *The Iliad*, keeping the original (75% / 15% / 15%) split. This allowed us to measure multi-domain learning ability. In this experiment we only compared the baseline RNN to the 2-layer LSTM.

Deeper models: For training deeper models we explored a 4-layer LSTM with and without dropout. When training with dropout we set the dropout rate at 0.3 for layers 1-3, with no dropout at the final layer. A change we had to make was to decrease the number of hidden neurons from the original 2-layer LSTM recipe down to 75, the other hyperparameters were kept the same.

5 Experiments

5.1 The impact of hidden sizes

The influence of hidden size is performed under other parameters remain unchanged. Three hidden sizes are tested within different model in this section: hidden-size=[10, 15, 20]. The fixed parameters for RNN are: seq length=25, batch size=16, num layers=1, num epochs=4, learning rate=0.0001725398929988799. The fixed parameters for 1 and 2 layers LSTM are: seq length=25, batch size=16, num layers=1 or 2, num epochs=4, learning rate=0.007. As refer to table 5, the table clearly states that the validation loss decreases when increases the size of hidden layer.

5.2 Hyperparameter search

From the hyperparameter search we found that the models in Table 1 gave the lowest validation loss. The 1-layer RNN is the baseline model that is used for comparisons. Train and validation loss refers to the lowest loss achieved during training. We did not investigate a 1-layer GRU in this study because the 2-layer GRU outperformed it.

Model	Hidden	Batch size	Learning rate	Train loss	Validation loss
1-layer RNN	306	28	0.0001725398929988799	1.54	1.85
1-layer LSTM	544	18	0.0007087765262043359	1.42	1.78
2-layer LSTM	576	64	0.004262771798835783	1.51	1.78
2-layer GRU	145	50	0.000845663544150917	1.45	1.79

Table 1: Validation and training loss for all models using the listed hyperparameter settings for training. Hidden refers to the number of neurons in the hidden layer.

From the result of the search in Table 1 there is no obvious trend among the models, except that a larger hidden size is preferred for the LSTM compared to GRU. Learning rate is also generally low for RNNs compared to image recognition models, which can also be seen in the Table. All four models achieve similar losses, but it seems like the baseline RNN loss is a bit higher than the rest.

5.3 Model evaluation

Table 2 shows the results of our evaluation metrics from the training with the optimal hyperparameters from Table 1. The models were saved at the best validation loss and then used for evaluation.

Model	Test loss	Perplexity	Spelling	Self-BLEU	BERT
1-layer RNN	1.94	2.94	94%	0.14	33%
1-layer LSTM	1.90	2.68	96%	0.14	35%
2-layer LSTM	1.89	2.68	94%	0.14	34%
2-layer GRU	1.88	2.83	93%	0.14	35%

Table 2: Evaluation metric results for all four models. Test loss was the loss on the test dataset and spelling represents the percentage of correctly spelled words according to our spellchecker.

Table 2 shows that the RNN had the highest test loss of all models by a small margin, indicating it generalized slightly worse than the others. This could be due to its limited ability to capture longer dependencies in sequence data. The LSTM’s memory capabilities could be one of the reasons why it scores lower on perplexity. The spelling percentages are relatively similar across all models. The self-BLEU score indicates that all models synthesize text with equal diversity. The BERT score is also similar for all models, suggesting that the outputs are equally semantically accurate. Quantitatively, the LSTM models show the best overall results by a small margin.

For all synthesized text, see Appendix A. As for the group’s human evaluation, it is difficult to draw a conclusion about which model performed best. We observed that all models had high similarity to Shakespearean style, characterized by dialogue and third-person descriptions. However, grammar and semantic coherence are low across all models, perhaps slightly worse in the RNN compared to the others. Occasionally, the model gets it right and produces interesting sentences, "CAMILLO: I know not speak it is the conscience is see the seall of the threater" - GRU. Sometimes not, "ROMEO: I this so much a children, sir," - RNN. Nucleus sampling worked well and produced less repetitive text, temperature experiments showed that 0.5 produced the best texts. Too low led to repetitive text, too high led to unreadable text.

5.4 Mixed dataset

Table 3 shows the perplexity score for the baseline RNN and 2-layer LSTM models on the three different datasets. The Shakespeare perplexity is the same as in earlier results. We used the same hyperparameters as written in Table 1 and trained each model for 20 epochs saving the model at the best validation loss.

Model	Shakespeare	The Iliad	Shakespeare + The Iliad
1-layer RNN	2.94	3.35	3.10
2-layer LSTM	2.68	2.48	2.51

Table 3: Perplexity score for two models on the specific datasets, rounded down.

Inspecting Table 3 it is clear that the 2-layer LSTM performs better than the RNN baseline on all three datasets. The training showed that the LSTM quickly overfitted the combined dataset, its best epoch was the third, while the baseline needed all epochs and was underfitted. The reason why *The Iliad* had higher perplexity than Shakespeare could be because it was more difficult to predict, or that Shakespeare enjoyed more training epochs. Also, the text synthesis differs between the two models when training on the combined dataset, the LSTM’s synthesis seems closer to a mix between Shakespeare and *The Iliad* than the baseline. The LSTM synthesized more repetitive and shorter sentences beginning with 'The' or 'And', compared to the RNN that synthesized longer more diverse sentences, however, the LSTM’s sentences showed an increase in Greek references. One interesting finding was that the synthesis seemed more distant from Shakespeare than *The Iliad*, but a reason for that could be because the models were evaluated on *The Iliad* alone. Even though it only made up 25% of the training data it still managed to capture parts of its features.

5.5 Deeper models and dropout layers

Table 4 shows the evaluation results from training a 4-layer LSTM model with and without dropout. The models were trained with the hyperparameters written in Table 1 and trained for 20 epochs saving the model at the best validation loss.

Model	Test loss	Perplexity	Spelling	Self-BLEU
4-layer LSTM	2.02	3.30	92%	0.14
4-layer LSTM w. dropout	2.29	4.35	75%	0.14

Table 4: Evaluation metric results for two models. Test loss was the loss on the test dataset and spelling represents the percentage of correctly spelled words according to our spellchecker.

Inspecting Table 4 it is clear that the dropout layers did not improve the model, performing worse on all tests, except for a similar BLEU score. We had to decrease the number of hidden neurons to keep losses down, larger models gave unpredictable text synthesis. Neither model achieved better results than the 2-layer LSTM, which could point to that increased layers are not improving performance, but another reason could be inadequate hyperparameters for deeper models. Additionally, we experimented with a 10-layer LSTM with and without dropout, but both had a significantly negative impact. The validation loss exceeded 3, and the model was unable to synthesize any meaningful text. This could be due to the model having too many layers, or it may indicate that the hidden size should have been reduced further.

6 Conclusion and discussion

To summarize our findings, it is possible to build models that synthesize text that relatively closely resembles the training data, and we observed performance differences between the models. The 1- and 2-layer LSTM models performed the best, followed by the 2-layer GRU, and finally the baseline 1-layer RNN model. One surprising observation was that the LSTM and GRU models did not significantly outperform the baseline RNN, despite their more complex architectures. Additionally, it is hard to compare the text synthesis between models, there is no clear winner writing the best text and they are equally good. It would be interesting to compare these models when trained with standard word embeddings to see if they are better able to handle poetry than the character-level encoding we used.

7 Limitations

Computational resources were limited, which constrained our ability to perform extensive hyperparameter searches, experiment with larger or deeper models, and fully explore multi-domain learning. We observed that the 2-layer LSTM model did not converge within a reasonable number of epochs. Due to the resource limitations, we implemented early stopping, if the validation loss did not improve within 3 consecutive epochs, training was stopped and the model with the lowest validation loss was saved. With access to more computational resources, more effective parameter search strategies, such as Bayesian optimization could be used. In addition, human evaluation of synthesized text quality is inherently subjective and time consuming, making it difficult to quantify reliably. More human review is needed for the quality of the text, enough but not limited to the aspects like semantic, coherent, grammar and Shakespeare similarity. The use of random seeds introduced reproducibility issues, especially for models with higher variability in training. Although we set seeds to control for randomness, results still varied slightly across runs due to stochastic components in training.

References

- [1] A. Sherstinsky, “Fundamentals of recurrent neural network (rnn) and long short-term memory (lstm) network,” *Physica D: Nonlinear Phenomena*, vol. 404, p. 132 306, 2020. DOI: <https://doi.org/10.1016/j.physd.2019.132306>. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0167278919305974>.
- [2] I. Sutskever, J. Martens, and G. E. Hinton, “Generating text with recurrent neural networks,” in *Proceedings of the 28th international conference on machine learning (ICML-11)*, 2011, pp. 1017–1024.
- [3] S. Mangal, P. Joshi, and R. Modak, “Lstm vs. gru vs. bidirectional rnn for script generation,” *arXiv preprint arXiv:1908.04332*, 2019.
- [4] “Bert-score: PyTorch implementation of BERT score.” version 0.3.13. (), [Online]. Available: https://github.com/Tiiiger/bert_score (visited on 05/16/2025).
- [5] T. Zhang, V. Kishore, F. Wu, K. Q. Weinberger, and Y. Artzi, *BERTScore: Evaluating text generation with BERT*, Feb. 24, 2020. DOI: 10.48550/arXiv.1904.09675. arXiv: 1904.09675[cs]. [Online]. Available: <http://arxiv.org/abs/1904.09675> (visited on 05/16/2025).
- [6] A. Holtzman, J. Buys, L. Du, M. Forbes, and Y. Choi, “The curious case of neural text degeneration,” *arXiv preprint arXiv:1904.09751*, 2019.
- [7] Y. Bengio, N. Boulanger-Lewandowski, and R. Pascanu, *Advances in optimizing recurrent networks*, Dec. 14, 2012. DOI: 10.48550/arXiv.1212.0901. arXiv: 1212.0901[cs]. [Online]. Available: <http://arxiv.org/abs/1212.0901> (visited on 05/18/2025).
- [8] “Pyspellchecker: Pure python spell checker based on work by peter norvig.” version 0.8.2. ().
- [9] Y. Zhu, S. Lu, L. Zheng, *et al.*, “Texygen: A benchmarking platform for text generation models,” in *The 41st International ACM SIGIR Conference on Research & Development in Information Retrieval*, ser. SIGIR ’18, New York, NY, USA: Association for Computing Machinery, Jun. 27, 2018, pp. 1097–1100, ISBN: 978-1-4503-5657-2. DOI: 10.1145/3209978.3210080. [Online]. Available: <https://dl.acm.org/doi/10.1145/3209978.3210080> (visited on 05/16/2025).

Appendix

A Generated text for each model

A.1 Reference text from Shakespeare

ROMEO: Alas, that love, whose view is muffled still, Should, without eyes, see pathways to his will! Where shall we dine? O me! What fray was here? Yet tell me not, for I have heard it all. Here's much to do with hate, but more with love. Why, then, O brawling love! O loving hate! O any thing, of nothing first create! O heavy lightness! serious vanity! Mis-shapen chaos of well-seeming forms! Feather of lead, bright smoke, cold fire, sick health! Still-waking sleep, that is not what it is! This love feel I, that feel no love in this. Dost thou not laugh?

BENVOLIO: No, coz, I rather weep.

ROMEO: Good heart, at what?

BENVOLIO: At thy good heart's oppression.

ROMEO: Why, such is love's transgression. Grievs of mine own lie heavy in my breast, Which thou wilt propagate, to have it prest With more of thine: this love that thou hast shown Doth add more grief to too much of mine own. Love is a smoke raised with the fume of sighs; Being purged, a fire sparkling in lovers' eyes; Being vex'd

A.2 1-layer RNN

ROMEO: I this so much a children, sir, the means the was the heaven and so but and be the some some some past our than the pardon them should not well thee to the brother to the sease of his fair our fair not and his lame: I would have shall be sing be sen to steel to me the grace him so he dead man the seater's poor to me, it is she the lest the hands the wars the shall be pertent the will and will shall be so many thee in the terth the trumpets, then made the ware stopport the sease your beather, that the offer is the best of thee, the fair and the part and in her his name: The great in the heavens of my tongue like and haste her and thee, the best of the sease your gracious partion of my call thee well so many that stay the scance in stand no less and hear stand of the hour sons as shall stand me, in the fair beat the prince one to my soul of the sease in his passes of the gall as my father, and the heart the bear the cause of the wife and be wear thee are to the may deserves and be gone.

A.3 1-layer LSTM

ROMEO: I am gone and see the strange of the land and the work of the honour, that will see thee the sea, I will be some patience to make us and the heart with the son, the good for the grown and angry to the son: what thou shalt see thee the prince to thee to be a seeming so much it the world and since the father next truth, of my prize it were some party the words of the woman: What say the sears and the pretty of the sentence more father to see him betwixt thee and like the son and been some part to make unto the heavens the sea how it be shame To make the bastard what you have see the word me with thee to me and the more Of the ship thee and the most a poor bed, the son: the singless a seesit of my prince of my ships are the mother shows what we are one my father than the cause of the winder to the world, the words of the singhis lamented of him, and the boy, and the man that be so and the good man with the man with the sing of my boy. LEONTES: I have some command thee to the business and

A.4 2-layer LSTM

ROMEO: London, and hear them all the bear and the three sons of the shepherd: The father will not stay to say the deed to save thee were point in the field of the singless and how to be the country: and then the shepherd how may be a single and the partience, here in the boy. A child the best of the shame and leave you my come to have seen speak thee with the prince more than thee, the gains, the next of the shame to me to hear with her the queen: so look the shame and for the great base shall be the queen, Whose best shepherd the man with the great pretty that was some part of all the time to the best shepherd: I would she does are some one shall be the prince the ship than the head of the next word the seasing and the soul in my lord, I would not received of the people of the child. 'Tis a

partious boy, but he shall be postimely weep and the while she shall be my thoughts of my life I have been seem thee to my soner of my best shame of the great day to the sea The prince a foolish they are

A.5 2-layer GRU

ROMEO: Deceived thee and be when the world from the seal the grace with a good dear the seath of the great be the common my followers, that we are the bear the present and see here be my county of my proceed to the golden the charged of the more well, then the blessing of my bear strength and she did be the presence the body my proceed the royal see the true shall be the shapen, if the hate when the but one more than the many foot shall be the serve me for the prince for the best of the shall not so like a heaven see the bost with stree, she did been be the sake made honour to the boying bear me not the father, be the winds, the which she may be death the world, be the shart were a great commender to the bound and the head, and let the such of the good father when the will of my farther than the time to the present be the prince better of the grace Which where is the great of the heart with heart in the gone.

CAMILLO: I know not speak it is the conscience is see the seall of the threatier, wh

A.6 1-layer RNN with nucleus sampling

"ROMEO: I will see the seall be the man of the sing, the sair heard make a say it well, and see how they see it one in the stity and with see her such as be not the this his majesty hands and things are this shall be well to the peace of the truth, by the send the world have and the fearful honour and the way and my son the hates be and be so my lord; The best and should be content.

PAULINA: I would speak, and make a still the peace, of the state of a foot are the charged the contion of the hearts to the bear thee for the holse to make her here a great and send the hands and the father before and such a man of the father's heart, to my soul'd to be the death.

POLIXENES: A heart a word of his grace it in the straight of mine and me on the great cannot be not to dead my lord; I think you are so my soul.

MERCUTIO: A thousand leave the such a consul to shall be made the near to see her heart.

LEONTES: The gods to make him and a pain of the worship the war here on the bed, so here sheep and my "

A.7 1-layer RNN from mixed dataset

we the son the bourder son the shall she to the doundsing the fields and forth her or the Greeks the grief and the goddens warrior spoke: The rock of partions he the state, and he all the sers death the sparing breast of the cares the forth here the flew from him sour in the shall the parts the the goddest the lows consent counted look of the fields and recone shower, And shall stand his say shall the deed, The gods the shand the winds the morning arms and the see she that slapost of the spared, and like the regenting the stands, Be mains the dishone the should the sain, and before the morn the spoke. The sounds the prence shall prove the shower their of the sparestor streath the rest the shance the faised where of the sugh the than the shame, And the farestrous shall the couch the rears of the show Greeks the shall streast the streams and the doum his stands of war to the force the sounder the read the shall the parts and the reathers the round, the tark of s all the batters to the fa

A.8 2-layer LSTM from mixed dataset

BOOK VII.

The dead and a forth the mother burns to the race, And Troy renown'd.

"O strength and heroes of the shoulders of the mournful arms the long to the Trojan sacred shade. The chief around, And the state of Trojan side; While should be seen the warrior of the sea-stands the earth of strong war, The state, and an end. The sense of Greece beneath the sons of Troy, Proud Menelaus were their brazen from the god of fall: The chief the strong strong inspired: "O should be sounds the spoil, The search the Greeks the boast a monster of the sense, and a sacred bowls late to

the fields raised the great Trojan steeds, and the mountain wrath, and the parts of arms and gods shall be heard, The bold monarch of the world: Ajax as the mains, and the shade of man: The corslet Ajax breathe with the shade. The business shall the pear when they wear our side of conquest was the foe. The ground the mother's sons of barred state.

Then thus is the powers and the spoil be every prove: The Greeks th

A.9 4-layer LSTM

ROMEO: it ere of you shall sheen speak be best the time of the shall the may not one, be pity headst they to be shest the been than what prove the bade your heads I seems of my good present the words: Here heart and thee and a them the boited how the forth, the persent me, and now the play the past in a see my words when the army nor betters in the grace the hand This linttent in the been life.

LEONTES: The see his brood grown the may be leave the is and his concent thought and the house then the see since this repence him, my revenge in my live the preated the was the see she the fair hence, and faint the now the thou as a life, make the can the dead, That preserty and heart the news and should say in this brown of the counter to the speed the thy seas that the time of thee to all me the shall some dead for the tie thy childing the head the persion.

PAULINA: The good liege: So the both instrenenest thou can the for the heads in the melt: And from we more from the good night of this for th

A.10 4-layer LSTM with Dropout

ROMEO: ene the by hear for and nor have by go hever, mes the be wise will here of he be pither the he ky to and beser the wear: On worg are, of the de hes the bore sore nounds ip sor the que wat the word hese Meed.

GAUNEN: And that so here the ween the merell the that met hes the the the shatl, hor for on the worg in feath bame the deatht shith the the he the and a pore the thit the and nhat to bere me worg sor his be hove whew the the buve nather is and with the frace in soand the hound then the so thit a thit in that hive thare bores the wer fhith the hears the hers the shear a father he a live a nor and nor ark word sither the the the care thee I the hear fear basped the cort nhee felver and say fhe nither the meath, Thit in that the what the you and the the of the cither! I wats for that, and the meatht, yearerer the with the the the eno the cise i math mather with of wert the wares the the lere thee hond thow the hor a a ceatelf, This the wear merert the the the rave word save the

B Size of hidden layer impact

Model	hidden size	Validation Loss
RNN	10	2.65
	15	2.54
	20	2.46
1-layer LSTM	10	2.24
	15	2.14
	20	2.09
2-layer LSTM	10	2.30
	15	2.21
	20	2.15

Table 5: Effect of hidden size across models